QUANTIUM VIRTUAL INTERNSHIP – TASK 1

Customer Segment & Chips Purchasing Behaviour Analysis

Strategic Recommendation Report for Julia, Category Manager

Prepared by: Asha Latha Maanepalli
Tools Used: MySQL, Excel, Power BI

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1. Executive Summary

This report analyses customer purchasing behaviour within the chips category using transaction and customer segment data.

The objective is to identify:

- Which customer segments drive category sales
- Which brands perform strongest
- What factors influence spending behaviour
- Commercial opportunities for growth

Key Outcomes

- **Kettle** is the highest-performing brand by revenue.
- **Older Singles/Couples** generate the highest overall category revenue.

- **Older Families** deliver the highest average spend per customer.
- **Young Singles/Couples (Mainstream)** represent the largest customer base.
- **175g packs** are the strongest-selling pack size.
- Different customer segments require different commercial strategies.

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2. Business Objective

Provide Julia with data-backed recommendations for the upcoming category review by understanding current customer behaviour and revenue drivers.

Focus Areas:

- Revenue by customer segment
- Brand preference by segment
- Pack size preference
- Spend per customer
- Growth opportunities

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3. Datasets Used

A. Transaction Dataset

Contains:

- Date
- Store Number
- Customer ID
- Product Name
- Quantity Purchased
- Total Sales

B. Purchase Behaviour Dataset

Contains:

- LIFESTAGE
- PREMIUM_CUSTOMER
- Customer ID

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4. Data Preparation & Cleaning

The following preparation steps were completed before analysis:

- Imported raw CSV files into MySQL
- Converted Excel serial dates into standard SQL date format

- Cleaned hidden characters and inconsistent text values
- Extracted pack size & Brand from product name
- Joined customer and transaction tables for final analysis

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5. Analytical Methods Used

- INNER JOIN
- GROUP BY
- Aggregations (SUM, AVG, COUNT DISTINCT)
- Subqueries
- Segment comparison
- Brand performance analysis
- Dashboard storytelling in Power BI

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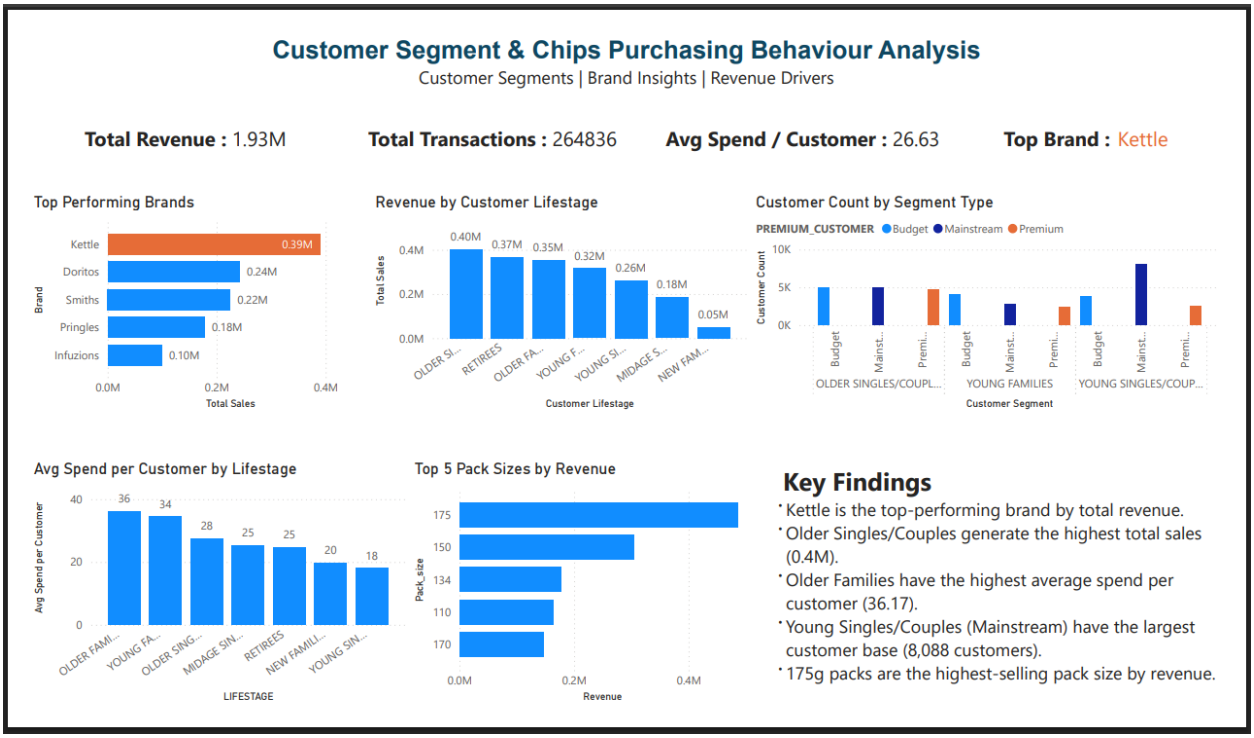
6. Dashboard KPI Summary

- **Total Revenue:** 1.93M
- **Total Transactions:** 264,836
- **Average Spend / Customer:** 26.63
- **Top Brand:** Kettle



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7. Key Findings

Finding 1: Kettle is the Leading Brand

Kettle generated the highest total revenue and consistently ranked #1 across multiple customer segments.

Top Brand Revenue:

1. Kettle – 390K+
2. Doritos – 241K+
3. Smiths – 224K+

Business Meaning

Kettle has strong cross-segment demand and premium brand strength.

Finding 2: Older Singles/Couples Drive Highest Total Revenue

This customer group generated the highest overall category sales.

Business Meaning

They are one of the most commercially valuable segments and should receive continued attention.

Finding 3: Older Families Have Highest Spend Per Customer

Average spend per customer reached approximately **36**, highest among all segments.

Kettle Metrics in Older Families:

- Avg Spend / Customer = 14.25
- Avg Quantity / Customer = 2.89
- Avg Price / Unit = 7.09

Business Meaning

This segment purchases more volume and accepts premium pricing.

Finding 4: Young Singles/Couples (Mainstream) Are a Growth Segment

Largest customer base:

- **8,088 customers**

Kettle Metrics:

- Avg Spend / Customer = 11.56
- Avg Quantity / Customer = 2.34
- Avg Price / Unit = 6.01

Business Meaning

Large population + premium purchasing behaviour = scalable growth segment.

Finding 5: 175g Packs Lead Category Revenue

Top performing pack sizes:

1. 175g
2. 150g
3. 134g

175g packs were highly preferred across major segments.

Business Meaning

This pack size should receive strongest shelf visibility and inventory priority.

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8. Strategic Recommendations

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1. Expand Kettle Brand Visibility

- Premium shelf placement
- Prevent stockouts
- Secondary display zones
- Bundle promotions

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2. Target Older Families

- Family-size bundles
- Multi-pack promotions
- Value messaging
- Grocery aisle visibility

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3. Grow Young Singles/Couples

- Premium branding campaigns

- Checkout placement
- Convenience-focused packs
- Digital/mobile promotions

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4. Optimize Pack Size Mix

Prioritize:

- 175g
- 150g
- 134g

Reduce low-performing sizes if shelf space is constrained.

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5. Use Secondary Brands Strategically

Leverage Doritos, Smiths and Pringles for:

- Variety strategy
- Cross-promotions
- Competitive shelf mix

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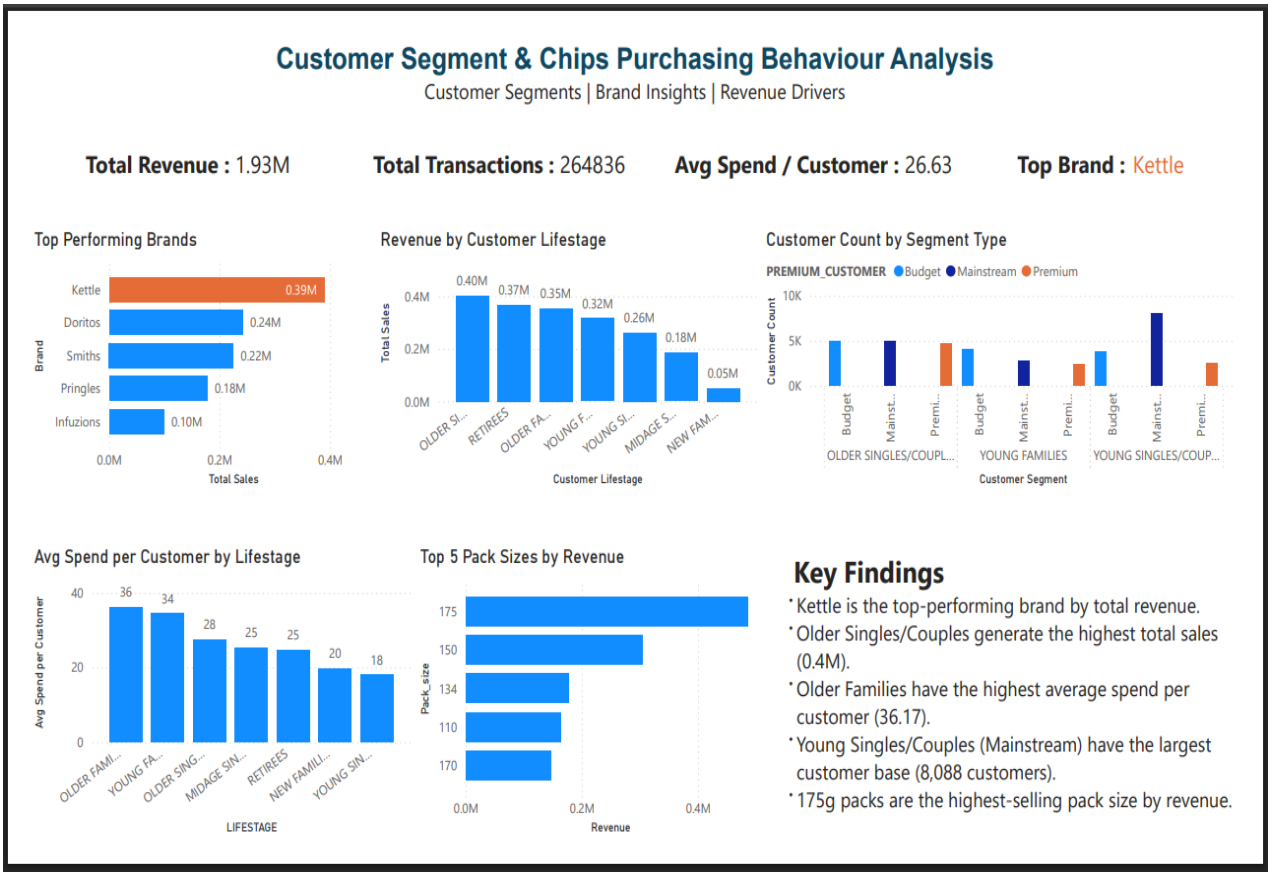
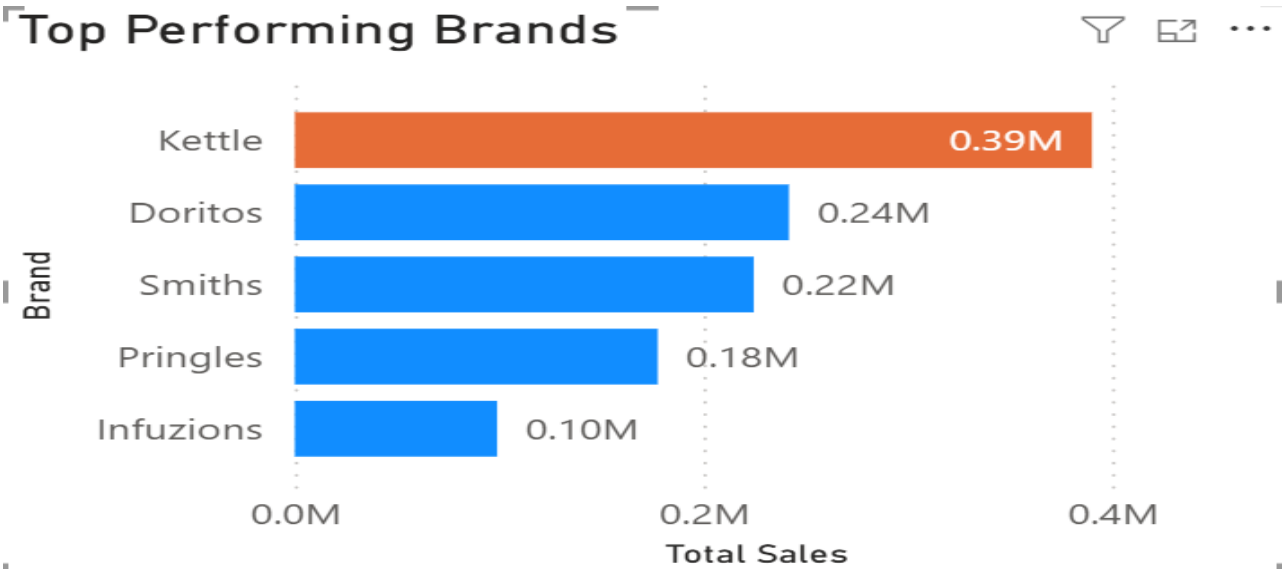
9. Dashboard Evidence


Customer Segment & Chips Purchasing Behaviour Analysis
Customer Segments | Brand Insights | Revenue Drivers
Total Revenue : 1.93M
Total Transactions : 264836
Avg Spend / Customer : 26.63
Top Brand : Kettle
Top Performing Brands
Kettle 0.39M
Doritos 0.24M
Smiths 0.22M
Pringles 0.18M
Infuzions 0.10M
Brand
0.0M 0.2M 0.4M
Total Sales
Revenue by Customer Lifestage
0.40M 0.37M 0.35M 0.32M 0.26M 0.18M 0.05M
Total Sales
0.0M 0.2M 0.4M
OLDER SI... RETIREES OLDER FA... YOUNG F... YOUNG SI... MIDAGE S... NEW FAM...
Customer Lifestage
Customer Count by Segment Type
PREMIUM_CUSTOMER Budget Mainstream Premium
Customer Count
10K 5K 0K
Budget Mainst... Premi... Budget Mainst... Premi... Budget Mainst... Premi...
OLDER SINGLES/COUPL... YOUNG FAMILIES YOUNG SINGLES/COUP...
Customer Segment
Avg Spend per Customer by Lifestage
Avg Spend per Customer
40 20 0
36 34 28 25 25 20 18
OLDER FAMI... YOUNG FA... OLDER SING... MIDAGE SIN... RETIREES NEW FAMILI... YOUNG SIN...
LIFESTAGE
Top 5 Pack Sizes by Revenue
Pack_size
175 150 134 110 170
0.0M 0.2M 0.4M
Revenue
Key Findings
Kettle is the top-performing brand by total revenue.
Older Singles/Couples generate the highest total sales (0.4M).
Older Families have the highest average spend per customer (36.17).
Young Singles/Couples (Mainstream) have the largest customer base (8,088 customers).
175g packs are the highest-selling pack size by revenue.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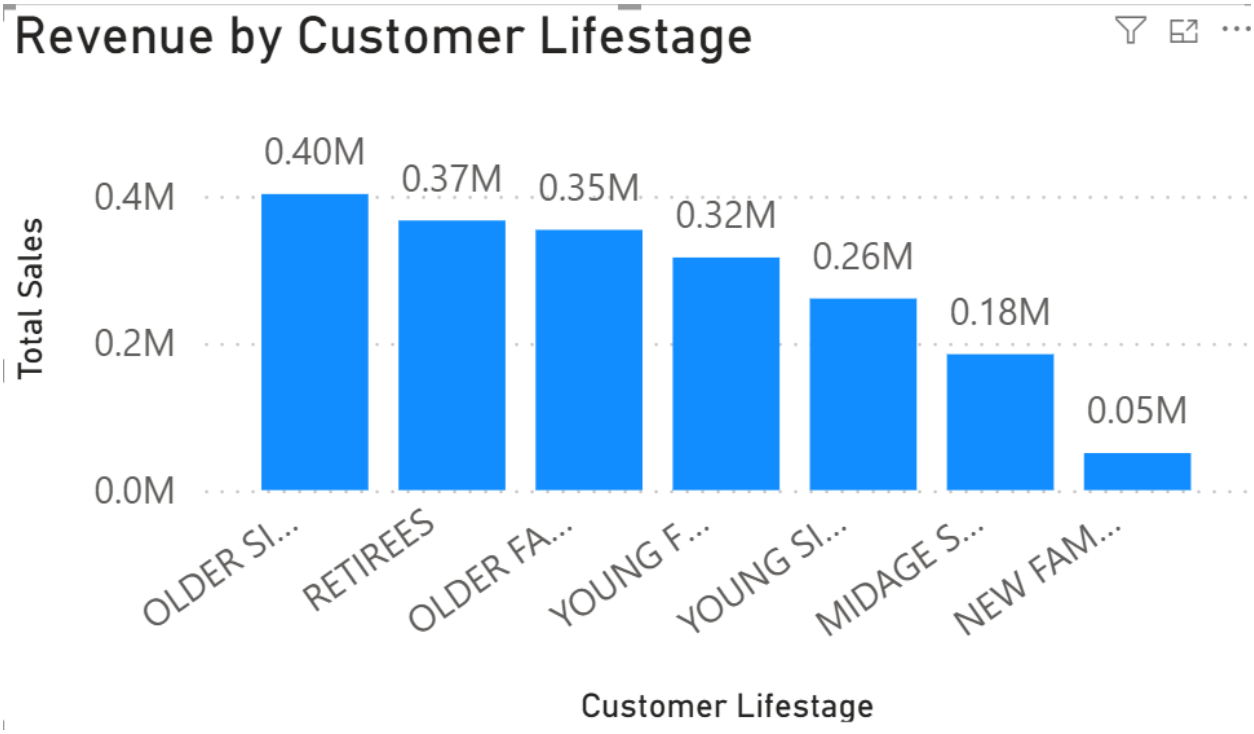
Revenue by Customer Lifestage
Total Sales
0.4M
0.2M
0.0M
0.40M
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OLDER SI...
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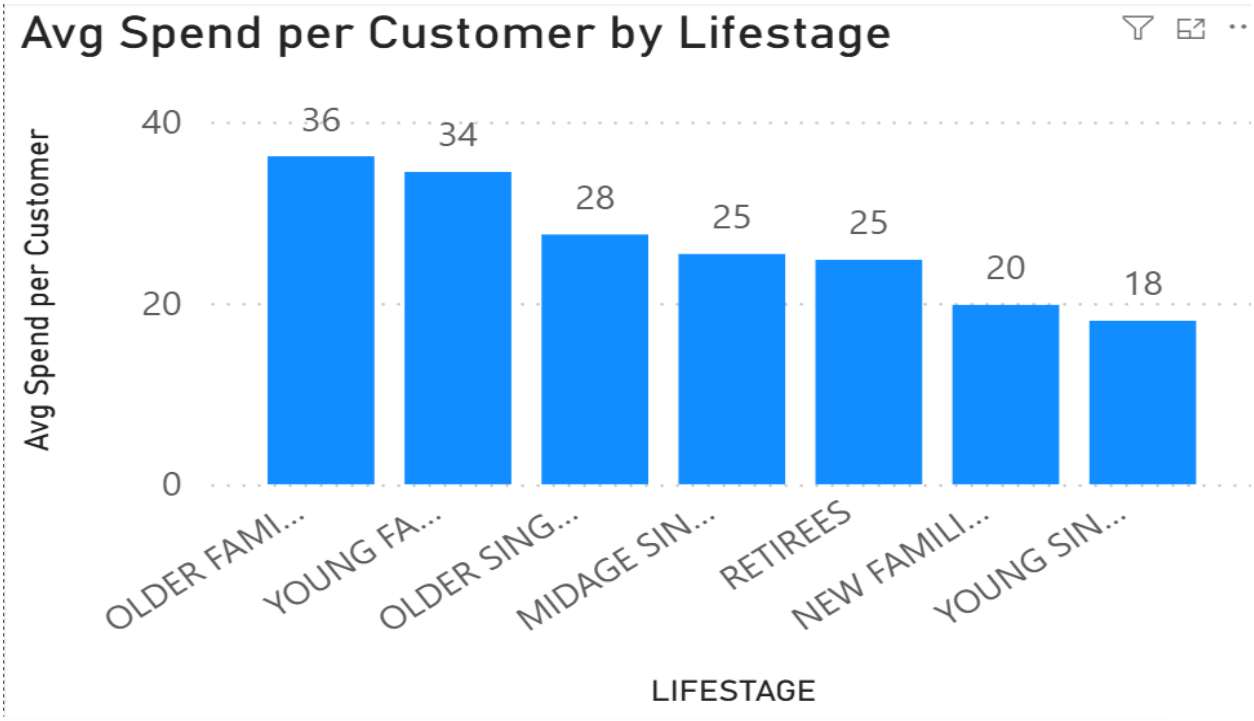
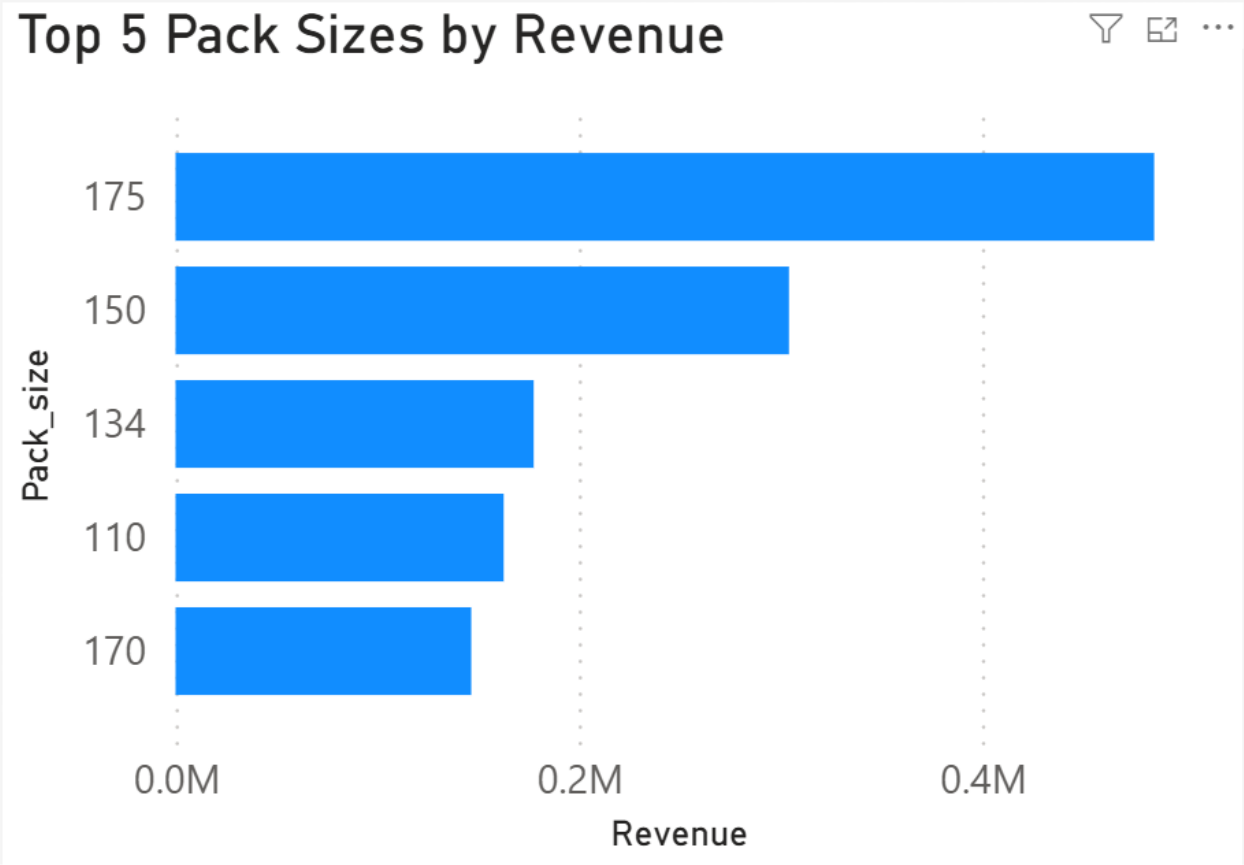
Customer Lifestage


Avg Spend per Customer by Lifestage
Avg Spend per Customer
40
20
0
36
34
28
25
25
20
18
OLDER FAMI...
YOUNG FA...
OLDER SING...
MIDAGE SIN...
RETIREES
NEW FAMILI...
YOUNG SIN...
LIFESTAGE



10. Final Recommendation to Julia

To improve chips category performance, Quantium should prioritize:

1. Kettle as hero brand
2. Older Families for value growth
3. Young Singles/Couples for premium growth
4. 175g packs as core size strategy
5. Data-led assortment and promotion planning

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11. SQL Queries Used

A. Data Preparation Queries

```sql
SET SQL_SAFE_UPDATES = 0;
ALTER TABLE transactions
ADD COLUMN txn_date DATE;
UPDATE transactions
SET txn_date = DATE_ADD('1899-12-30', INTERVAL EXCEL_DATES DAY);
SET SQL_SAFE_UPDATES = 1;
SELECT txn_date FROM transactions LIMIT 5;

ALTER TABLE transactions
ADD COLUMN pack_size_g INT;
UPDATE transactions
SET pack_size_g =
CAST(
REGEXP_SUBSTR(PROD_NAME, '[0-9]+(?=g)')
AS UNSIGNED
);
SELECT PROD_NAME, pack_size_g
FROM transactions
LIMIT 10;
```

B. Customer Segment Revenue Analysis

```sql
SELECT p.LIFESTAGE,
       ROUND(SUM(t.TOT_SALES),2) AS total_sales
FROM purchase_behaviour p
JOIN transactions t
ON p.LYLTY_CARD_NBR = t.LYLTY_CARD_NBR
GROUP BY p.LIFESTAGE
ORDER BY total_sales DESC;
```

C. Brand Revenue Ranking

```sql
select brand,round(sum(tot_sales),2) as total_sales
from transactions
group by brand
order by total_sales desc;
```

D. Average Customer Spent By Lifestage

```sql
SELECT LIFESTAGE,
       AVG(customer_total) AS avg_customer_spend
FROM (
    SELECT t.LYLTY_CARD_NBR,
           p.LIFESTAGE,
           SUM(t.TOT_SALES) AS customer_total
    FROM transactions t
    JOIN purchase_behaviour p
    ON t.LYLTY_CARD_NBR = p.LYLTY_CARD_NBR
    GROUP BY t.LYLTY_CARD_NBR, p.LIFESTAGE
) x
GROUP BY LIFESTAGE
ORDER BY avg_customer_spend desc;
```

12. Conclusion

Customer segmentation strongly influences purchasing behaviour.

Kettle emerged as the strongest brand, while Older Families and Young Singles/Couples represent high-opportunity segments for different commercial reasons.

These findings can directly support pricing, merchandising, promotional planning and future category growth decisions.

Prepared using MySQL, Power BI & Business Analytics Thinking